MA1522 Linear Algebra for Computing
L2 Lesson 9



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Recall that: Subspaces

Theorem

Let V be a subset of $\mathbb{R}^n$. The following statements are equivalent:

1. $V = \text{span}(S)$ *for some (finite) subset S of $\mathbb{R}^n$.*
2. $\mathbf{0}_n \in V$ *and for all $\boldsymbol{v}_1, \boldsymbol{v}_2 \in V$ and $c \in \mathbb{R}$, $\boldsymbol{v}_1 + \boldsymbol{v}_2 \in V$ and $c\boldsymbol{v}_1 \in V$.*
3. $\mathbf{0}_n \in V$ *and for all $\boldsymbol{v}_1, \boldsymbol{v}_2 \in V$ and $c_1, c_2 \in \mathbb{R}$, $c_1\boldsymbol{v}_1 + c_2\boldsymbol{v}_2 \in V$.*
4. $V = \{\boldsymbol{x} \in \mathbb{R}^n \mid \boldsymbol{A}\boldsymbol{x} = \mathbf{0}_m\}$ *for some $m \in \mathbb{Z}^+$ and $m \times n$ matrix $\boldsymbol{A}$.*

When any of the above statements hold, we say that V is a **(vector) subspace** *of $\mathbb{R}^n$.*

Theorem

If V and W are subspaces of $\mathbb{R}^n$, then $V \cup W$ is a subspace only if $V \subseteq W$ or $W \subseteq V$.

Question 1

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If U and W are subspaces of $\mathbb{R}^n$, is it (always) true that $U \cap W$ is a subspace of $\mathbb{R}^n$? Justify your answer.

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- Let $\boldsymbol{v}_1, \boldsymbol{v}_2 \in U \cap W$ and $c \in \mathbb{R}$. Then $\boldsymbol{v}_1, \boldsymbol{v}_2 \in U$ and $\boldsymbol{v}_1, \boldsymbol{v}_2 \in W$.

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  - Thus $\boldsymbol{v}_1 + \boldsymbol{v}_2 \in U$ (by closure in U) and $\boldsymbol{v}_1 + \boldsymbol{v}_2 \in W$ (by closure in W), and hence $\boldsymbol{v}_1 + \boldsymbol{v}_2 \in U \cap W$.

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  - Also, $c\boldsymbol{v}_1 \in U$ (by closure in U) and $c\boldsymbol{v}_1 \in W$ (by closure in W), and hence $c\boldsymbol{v}_1 \in U \cap W$.

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  - Also, $c\boldsymbol{v}_1 \in U$ (by closure in U) and $c\boldsymbol{v}_1 \in W$ (by closure in W), and hence $c\boldsymbol{v}_1 \in U \cap W$.

Thus $U \cap W$ is always a subspace. □

Computing $U \cap W$

Let $U = \text{span}\{\boldsymbol{u}_1, \ldots, \boldsymbol{u}_k\}$ and $W = \text{span}\{\boldsymbol{w}_1, \ldots, \boldsymbol{w}_l\}$ be subspaces of $\mathbb{R}^n$.

Then

$$\begin{aligned}
\boldsymbol{v} \in U \cap W &\Leftrightarrow \boldsymbol{v} \in U \text{ and } \boldsymbol{v} \in W \\
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&\Leftrightarrow \boldsymbol{v} = c_1\boldsymbol{u}_1 + \cdots + c_k\boldsymbol{u}_k = d_1\boldsymbol{w}_1 + \cdots + d_l\boldsymbol{w}_l \\
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and $\boldsymbol{v} = c_1\boldsymbol{u}_1 + \cdots + c_k\boldsymbol{u}_k$.

Thus we need to solve for $c_1, \ldots, c_k$ satisfying the last matrix equation.

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Question 2

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Let $U = \text{span}\{\boldsymbol{u}_1, \boldsymbol{u}_2\}$ and $W = \text{span}\{\boldsymbol{w}_1, \boldsymbol{w}_2\}$, where

$$\boldsymbol{u}_1 = (1,1,1,1), \quad \boldsymbol{u}_2 = (1,2,2,1), \quad \boldsymbol{w}_1 = (1,0,1,0), \quad \boldsymbol{w}_2 = (1,0,2,-1).$$

Find a subset S of $\mathbb{R}^4$ such that $U \cap W = \text{span}(S)$.

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Solution.

$$\begin{pmatrix} \boldsymbol{u}_1 & \boldsymbol{u}_2 & \boldsymbol{w}_1 & \boldsymbol{w}_2 \end{pmatrix} \xrightarrow[\text{Jordan}]{\text{Gauss-}} \begin{pmatrix} 1 & 0 & 0 & -2 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 2 \\ 0 & 0 & 0 & 0 \end{pmatrix}.$$

Thus $c_1 = 2t$, $c_2 = -t$, $-d_1 = -2t$, $-d_2 = t$, $t \in \mathbb{R}$. Hence

$$\begin{aligned} U \cap W &= \{c_1\boldsymbol{u}_1 + c_2\boldsymbol{u}_2\} \\ &= \{2t(1,1,1,1) - t(1,2,2,1) \mid t \in \mathbb{R}\} \\ &= \{t(1,0,0,1) \mid t \in \mathbb{R}\} = \text{span}\{(1,0,0,1)\}. \end{aligned}$$



Linear Independence

Let $S = \{\boldsymbol{v}_1, \ldots, \boldsymbol{v}_k\} \subseteq \mathbb{R}^n$. Let $\boldsymbol{A} = \begin{pmatrix} \boldsymbol{v}_1 & \boldsymbol{v}_2 & \cdots & \boldsymbol{v}_k \end{pmatrix}$.

Then

S is **linearly independent**

$\Leftrightarrow (c_1\boldsymbol{v}_1 + c_2\boldsymbol{v}_2 + \cdots + c_k\boldsymbol{v}_k = \boldsymbol{0} \Rightarrow c_1 = c_2 = \cdots = c_k = 0)$

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Theorem

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   a. S *is linearly* **dependent** *if*
      - $\mathbf{0} \in S$, *or*
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      - $\boldsymbol{v}_i \in \operatorname{span}\{\boldsymbol{v}_1, \ldots, \boldsymbol{v}_{i-1}, \boldsymbol{v}_{i+1}, \ldots, \boldsymbol{v}_k\}$ *for some* i.

   b. *Suppose that* S *is linearly independent.*
      i. *If* $T \subseteq S$, *then* T *is linearly independent.*
      ii. *If* $\boldsymbol{v} \in \mathbb{R}^n \setminus \operatorname{span}(S)$, *then* $S \cup \{\boldsymbol{v}\}$ *is linearly independent.*

2. $\{\boldsymbol{v}\}$ *is linearly independent* $\Leftrightarrow \boldsymbol{v} \neq \mathbf{0}$.

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      - $\boldsymbol{v}_i \in \operatorname{span}\{\boldsymbol{v}_1, \ldots, \boldsymbol{v}_{i-1}, \boldsymbol{v}_{i+1}, \ldots, \boldsymbol{v}_k\}$ *for some* i.
   b. *Suppose that* S *is linearly independent.*
      i. *If* $T \subseteq S$, *then* T *is linearly independent.*
      ii. *If* $\boldsymbol{v} \in \mathbb{R}^n \setminus \operatorname{span}(S)$, *then* $S \cup \{\boldsymbol{v}\}$ *is linearly independent.*
2. $\{\boldsymbol{v}\}$ *is linearly independent* $\Leftrightarrow \boldsymbol{v} \neq \mathbf{0}$.
3. $\{\boldsymbol{v}_1, \boldsymbol{v}_2\}$ *is linearly independent* $\Leftrightarrow \boldsymbol{v}_1, \boldsymbol{v}_2 \neq \mathbf{0}$ *and* $\boldsymbol{v}_1 \neq c\boldsymbol{v}_2$ *for all* $c \in \mathbb{R}$.

Question 3

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Determine if each of the following sets is linearly independent:

a. $\{(2,-1,0),(0,3,2),(2,4,3),(3,6,6)\}$.

b. $\{(1,1,0),(3,4,2)\}$.

c. $\{(1,1,0),(3,4,2),(0,0,0)\}$.

d. $\{(1,0,0),(0,1,1),(1,2,-1)\}$.

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Determine if each of the following sets is linearly independent:

a. $\{(2,-1,0),(0,3,2),(2,4,3),(3,6,6)\}$. ..... **Ans: False**

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For (a), find a subset S that is linearly independent and express the vector(s) not in S as a linear combination of the vector(s) in S.

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For (a), find a subset S that is linearly independent and express the vector(s) not in S as a linear combination of the vector(s) in S.

Solution.

$$\text{RREF}\begin{pmatrix} 2 & 0 & 2 & 3 \\ -1 & 3 & 4 & 6 \\ 0 & 2 & 3 & 6 \end{pmatrix} = \begin{pmatrix} 1 & 0 & 0 & \frac{9}{2} \\ 0 & 1 & 0 & \frac{15}{2} \\ 0 & 0 & 1 & -3 \end{pmatrix}.$$

Thus $\{(2,-1,0),(0,3,2),(2,4,3)\}$ is linearly independent, and

$$(3,6,6) = \tfrac{9}{2}(2,-1,0) + \tfrac{15}{2}(0,3,2) - 3(2,4,3).$$

□

Next Lesson ...

We will explore **Sections 3.6 and 3.7**.
Please watch/review the video clips for these sections before then.